

2.3 Properties of Determinants and Cramer's Rule

1. $\det(kA) = k^n \det(A)$

$$\begin{vmatrix} ka_{11} & ka_{12} & ka_{13} \\ ka_{21} & ka_{22} & ka_{23} \\ ka_{31} & ka_{32} & ka_{33} \end{vmatrix} = k^3 \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

2. $\det(A^{-1}) = \frac{1}{\det(A)}$

6. $\det(AB) = \det(A)\det(B)$

↳ If A and B are

3. $\det(A^T) = \det(A)$

square matrix of

4. $\det(A^k) = (\det(A))^k$

same size.

5. A is invertible iff $\det(A) \neq 0$

Example: 1 $\det(A+B) \neq \det(A) + \det(B)$

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

$$\det(A+B)$$

$$\det(A) + \det(B)$$

$$= \det \left(\begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} + \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \right)$$

$$\det(A) = \det \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$$

$$= \det \begin{bmatrix} 4 & 3 \\ 3 & 8 \end{bmatrix}$$

$$\det(A) = 1$$

$$\det(B) = 8$$

$$= \det 23$$

$$\det(A) + \det(B) = 9$$

Hence $\det(A+B) \neq \det(A) + \det(B)$

Example: 2 (Sum of Determinants)

$$\det \begin{bmatrix} 1 & 7 & 5 \\ 2 & 0 & 3 \\ 1 & 0 & 4 \end{bmatrix} = \det \begin{bmatrix} 1 & 7 & 5 \\ 2 & 0 & 3 \\ 1 & 4 & 7 \end{bmatrix} + \det \begin{bmatrix} 1 & 7 & 5 \\ 2 & 0 & 3 \\ 0 & 1 & 1 \end{bmatrix}$$

Simple Evaluate LHS and
then RHS

Example: 3 (Determinant For Invertibility)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 4 & 6 \end{bmatrix}$$

Since R_1 & R_3 are proportional so
 $\det(A) = 0$ Hence inverse not exist.

Example: 4 Verify $\det(AB) = \det(A)\det(B)$

$$A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}, B = \begin{bmatrix} -1 & 3 \\ 5 & 8 \end{bmatrix}, AB = \begin{bmatrix} 2 & 17 \\ 3 & 14 \end{bmatrix}$$

$$\det(AB):-$$

$$\det(AB) = \begin{bmatrix} 2 & 17 \\ 3 & 14 \end{bmatrix}$$

$$\det(AB) = 28 - 51$$

$$\det(AB) = -23$$

$$\det(A) \det(B):-$$

$$\det(A) = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

$$\det(A) = 1$$

$$\det(B) = \begin{bmatrix} -1 & 3 \\ 5 & 8 \end{bmatrix}$$

$$\det(B) = -8 - 15$$

$$\det(B) = -23$$

$$\det(A) \det(B) = (1)(-23) = -23$$

Hence LHS = RHS.

Example : 5

$$A = \begin{bmatrix} 3 & 2 & -1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{bmatrix}$$

$$C_{11} = 12, C_{12} = 6, C_{13} = -16$$

$$C_{21} = 4, C_{22} = 2, C_{23} = 16$$

$$C_{31} = 12, C_{32} = -10, C_{33} = 16$$

Same Row with Same Cofactors

Cofactor expansion of $\det(A)$

along First row:

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

$$\det(A) = 3(12) + 2(6) + (-1)(-16)$$

$$\det(A) = 36 + 12 + 16$$

$$\det(A) = 64$$

Along First Column

$$\det(A) = a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31}$$

$$\det(A) = 3(12) + 1(4) + (2)(12)$$

$$\det(A) = 36 + 4 + 24$$

$$\det(A) = 64$$

Different Row with Different Cofactors

$$\det(A) = a_{11}C_{21} + a_{12}C_{22} + a_{13}C_{23}$$

$$\det(A) = 3(4) + 2(2) + (-1)(16)$$

$$\det(A) = 12 + 4 - 16$$

$$\det(A) = 0$$

Conclusion:

This example demonstrate expanding matrix with its own cofactors yield unique determ (64) whereas multiplying row or column entries by different row or columns cofactors always result in 0.

Example: 6 (Adjoint of 3×3 Matrix)

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{bmatrix}$$

same matrix as of example 5 & its cofactors are given in example 5. put cofactors

$$A = \begin{bmatrix} 12 & 6 & -16 \\ 4 & 2 & 16 \\ 12 & -10 & 16 \end{bmatrix}$$

Take Transpose

$$\text{adj}(A) = \begin{bmatrix} 12 & 4 & 12 \\ 6 & 2 & -10 \\ -16 & 16 & 16 \end{bmatrix}$$

Example: 7 (use adjoint to find an inverse)
extension of example 6 & 5.

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

$$A^{-1} = \frac{1}{64} \begin{bmatrix} 12 & 4 & 12 \\ 6 & 2 & -10 \\ -16 & 16 & 16 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{3}{16} & \frac{1}{16} & \frac{3}{16} \\ \frac{3}{32} & \frac{1}{32} & -\frac{5}{32} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{bmatrix}$$

Ans

Example: 8 (using Cramer's rule to solve linear system)

$$\begin{aligned} x_1 + 2x_3 &= 6 \\ -3x_1 + 4x_2 + 6x_3 &= 30 \\ -x_1 - 2x_2 + 3x_3 &= 8 \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & 2 \\ -3 & 4 & 6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 30 \\ 8 \end{bmatrix}$$

A X B

$$X = A^{-1} B$$

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -3 & 4 & 6 \\ -1 & -2 & 3 \end{bmatrix}, \quad A_1 = \begin{bmatrix} 6 & 0 & 2 \\ 30 & 4 & 6 \\ 8 & -2 & 3 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 1 & 6 & 2 \\ -3 & 30 & 6 \\ -1 & 8 & 3 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 1 & 0 & 6 \\ -3 & 4 & 30 \\ -1 & -2 & 8 \end{bmatrix}$$

$$x_1 = \frac{\det(A_1)}{\det(A)}$$

$$x_2 = \frac{\det(A_2)}{\det(A)}$$

$$x_3 = \frac{\det(A_3)}{\det(A)}$$

$$x_1 = \frac{-40}{44}$$

$$x_2 = \frac{72}{44}$$

$$x_3 = \frac{152}{44}$$

$$x_1 = \frac{-10}{11}$$

$$x_2 = \frac{18}{11}$$

$$x_3 = \frac{38}{11}$$

$$(x_1, x_2, x_3) = \left(\frac{-10}{11}, \frac{18}{11}, \frac{38}{11} \right)$$

we will further solve exercise 98 in detail to check how we have find these determinants

Exercise: 2-3

Question: 1-4

verify that $\det(kA) = k^n \det(A)$

1-

$$A = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}, k=2$$

$$\det(kA) = k^n \det(A)$$

$$\text{L.H.S.} = \det(kA)$$

$$\text{R.H.S.} = k^n \det(A)$$

$$= \det \left(2 \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix} \right)$$

$$= 2^2 \det \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$= \det \begin{bmatrix} -2 & 4 \\ 6 & 8 \end{bmatrix}$$

$$= 4(-4-6)$$

$$= 4(-10)$$

$$= -40$$

$$= -40$$

$$\text{Hence } \det(kA) = k^n \det(A)$$

3-

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 3 & 2 & 1 \\ 1 & 4 & 5 \end{bmatrix}, k=-2$$

$$\text{L.H.S} = \det(kA)$$

$$= \det \left(-2 \begin{bmatrix} 2 & -1 & 3 \\ 3 & 2 & 1 \\ 1 & 4 & 5 \end{bmatrix} \right) \rightarrow \det \begin{bmatrix} -4 & 2 & -6 \\ -6 & -4 & -2 \\ -2 & -8 & -10 \end{bmatrix}$$

Expand along 1st Row.

$$= -4 \begin{vmatrix} -4 & -2 \\ -8 & -10 \end{vmatrix} - 2 \begin{vmatrix} -6 & -2 \\ -2 & -10 \end{vmatrix} + (-6) \begin{vmatrix} -6 & -4 \\ -2 & -8 \end{vmatrix}$$

$$= -4(40 - 16) - 2(60 - 4) - 6(48 - 8)$$

$$= -4(24) - 2(56) - 240$$

$$= -96 - 112 - 240$$

$$= -448$$

$$\text{R.H.S} = k^n \det(A)$$

$$= (-2)^3 \det \begin{bmatrix} 2 & -1 & 3 \\ 3 & 2 & 1 \\ 1 & 4 & 5 \end{bmatrix}$$

$$= -8 \left[2 \begin{vmatrix} 2 & 1 \\ 4 & 5 \end{vmatrix} - (-1) \begin{vmatrix} 3 & 1 \\ 1 & 5 \end{vmatrix} + 3 \begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix} \right]$$

$$= -8 \left[2(10 - 4) + 1(15 - 1) + 3(12 - 2) \right]$$

$$= -8(12 + 14 + 30)$$

$$= -448$$

Hence verified.

2 and 4

same as part 1 & 3

Question: 5-6

verify that $\det(AB) = \det(BA)$ and determine whether equality $\det(A+B)$ and $\det(A) + \det(B)$ holds.

6-

$$A = \begin{bmatrix} -1 & 8 & 2 \\ 1 & 0 & -1 \\ -2 & 2 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 2 & -1 & 4 \\ 1 & 1 & 3 \\ 0 & 3 & -1 \end{bmatrix}$$

$$\det(AB) = \det(BA)$$

$$\det(A+B) = \det(A) + \det(B)$$

5-

Follow the procedure

if LHS = RHS equality

both are
same

Holds otherwise not.

Exercise: 7-14

Use determinants to decide whether given matrix is invertible:

9-

$$A = \begin{vmatrix} 2 & -3 & 5 \\ 0 & 1 & -3 \\ 0 & 0 & 2 \end{vmatrix}$$

$$\det(A) = \begin{vmatrix} 2 & -3 & 5 \\ 0 & 1 & -3 \\ 0 & 0 & 2 \end{vmatrix}$$

Expand along 3rd row.

$$\det(A) = 0 + 0 + (-1)^{3+3} (2) \begin{vmatrix} 2 & -3 \\ 0 & 1 \end{vmatrix}$$

$$\det(A) = 2(2-0)$$

$$\det(A) = 4$$

As $\det(A) \neq 0$ so A is invertible

14-

$$A = \begin{vmatrix} \sqrt{2} & -\sqrt{7} & 0 \\ 3\sqrt{2} & -3\sqrt{7} & 0 \\ 5 & -9 & 0 \end{vmatrix}$$

as c_3 is zero so $\det(A) = 0$
Hence A is not invertible.

7-8-10-11-12-13

Use their respective metrics to find determinant. If determ. turn out to be non zero then matrix is invertible and if 0 then not invertible.

Question: 15-18

Find values of k for which matrix A is invertible.

15-

$$A = \begin{bmatrix} k-3 & -2 \\ -2 & k-2 \end{bmatrix}$$

$$\det(A) = (k-3)(k-2) - (-2)(-2)$$

$$\det(A) = k^2 - 2k - 3k + 6 - 4$$

$$\det(A) = k^2 - 5k + 2$$

$$\text{set } \det(A) = 0$$

$$0 = k^2 - 5k + 2$$

$$\Rightarrow k^2 - 5k + 2 = 0$$

$$a=1, b=-5, c=2$$

$$k = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{5 \pm \sqrt{25 - 8}}{2}$$

$$k = \frac{5 \pm \sqrt{17}}{2}$$

So A is invertible for all values of k
except $k = \frac{5 \pm \sqrt{17}}{2}$

18-

$$A = \begin{vmatrix} 1 & 2 & 0 \\ k & 1 & k \\ 0 & 2 & 1 \end{vmatrix}$$

$$\det(A) = \begin{vmatrix} 1 & 2 & 0 \\ k & 1 & k \\ 0 & 2 & 1 \end{vmatrix}$$

$$\det(A) = 1 \begin{vmatrix} 1 & k \\ 2 & 1 \end{vmatrix} - 2 \begin{vmatrix} k & k \\ 0 & 1 \end{vmatrix} + 0$$

$$\det(A) = (1 - 2k) - 2(k - 0) + 0$$

$$\det(A) = 1 - 2k - 2k$$

$$\det(A) = 1 - 4k$$

$$0 = 1 - 4k$$

$$4k = 1 \quad k = \frac{1}{4}$$

$$k \in \mathbb{R} - \left\{ \frac{1}{4} \right\}$$

k belongs to all real numbers except

$$k = \frac{1}{4}$$

16-17

same method followed.

Question : 19-23 : Decide A is invertible if so use adjoint method to solve for inverse.

23-

$$A = \begin{bmatrix} 1 & 3 & 1 & 1 \\ 2 & 5 & 2 & 2 \\ 1 & 3 & 8 & 9 \\ 1 & 3 & 2 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \text{adj } A$$

$\det(A) = ?$

$$\det(A) = \begin{vmatrix} 1 & 3 & 1 & 1 \\ 2 & 5 & 2 & 2 \\ 1 & 3 & 8 & 9 \\ 1 & 3 & 2 & 2 \end{vmatrix}$$

$$\det(A) = \begin{vmatrix} 1 & 3 & 1 & 1 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 7 & 8 \\ 0 & 0 & 1 & 1 \end{vmatrix} \begin{array}{l} R_2 - 2R_1 \\ R_3 - R_1 \\ R_4 - R_1 \end{array}$$

Expand Along 1st column.

$$\det(A) = 1 \begin{vmatrix} -1 & 0 & 0 \\ 0 & 7 & 8 \\ 0 & 1 & 1 \end{vmatrix} + 0 + 0 + 0$$

expand again along C_1

$$\det(A) = (1)(-1) \begin{vmatrix} 7 & 8 \\ 1 & 1 \end{vmatrix} + 0 + 0$$

$$\det(A) = 1 \left[(-1)(7-8) \right]$$

$$\det(A) = 1$$

$$\text{adj}(A) = ?$$

Find Minors :-

$A =$

$$\begin{bmatrix} 1 & 3 & 1 & 1 \\ 2 & 5 & 2 & 2 \\ 1 & 3 & 8 & 9 \\ 1 & 3 & 2 & 2 \end{bmatrix}$$

For a_{11} :-

$$M_{11} = 1 \begin{vmatrix} 5 & 2 & 2 \\ 3 & 8 & 9 \\ 3 & 2 & 2 \end{vmatrix}$$

PROBLEM

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$$M_1 = 5 \begin{vmatrix} 8 & 9 \\ 2 & 2 \end{vmatrix} - 2 \begin{vmatrix} 3 & 9 \\ 3 & 2 \end{vmatrix} + 2 \begin{vmatrix} 3 & 2 \\ 3 & 2 \end{vmatrix}$$

$$M_1 = 5(16 - 18) - 2(6 - 27) + 2(4 - 24)$$

$$M_1 = 5(-2) - 2(-21) + 2(-18)$$

$$M_1 = -10 + 42 - 36$$

$$M_1 = -4$$

For a_2

$$M_{12} = \begin{vmatrix} 2 & 2 & 2 \\ 1 & 8 & 9 \\ 1 & 2 & 2 \end{vmatrix}$$

$$M_{12} = 2 \begin{vmatrix} 8 & 9 \\ 2 & 2 \end{vmatrix} - 2 \begin{vmatrix} 1 & 9 \\ 1 & 2 \end{vmatrix} + 2 \begin{vmatrix} 1 & 8 \\ 1 & 2 \end{vmatrix}$$

$$M_{12} = 2(16 - 18) - 2(2 - 9) + 2(2 - 8)$$

$$M_{12} = 2(-2) - 2(-7) + 2(-6)$$

$$M_{12} = -4 + 14 - 12$$

$$M_{12} = -2$$

For a_3

$$M_{13} = \begin{vmatrix} 2 & 5 & 2 \\ 1 & 3 & 9 \\ 1 & 3 & 2 \end{vmatrix}$$

$$M_{13} = 2 \begin{vmatrix} 3 & 9 \\ 3 & 2 \end{vmatrix} - 5 \begin{vmatrix} 1 & 9 \\ 1 & 2 \end{vmatrix} + 2 \begin{vmatrix} 1 & 3 \\ 1 & 3 \end{vmatrix}$$

$$M_{13} = 2(6-27) - 5(2-9) + 2(3-3)$$

$$M_{13} = -42 + 35 + 0$$

$$M_{13} = -7$$

For a_{14}

$$M_{14} = -6$$

For a_{21}

$$M_{21} = -3$$

For a_{22}

$$M_{22} = -1$$

For a_{23}

$$M_{23} = 0$$

For a_{24}

$$M_{24} = 0$$

For a_{31}

$$M_{31} = 0$$

For a_{32}

$$M_{32} = 0$$

For a_{33}

$$M_{33} = -1$$

For a_{34}

$$M_{34} = -1$$

 a_{41}

$$M_{41} = +1$$

For a_{12}

$$M_{12} = 0$$

For a_{13}

$$M_{13} = -8$$

For a_{14}

$$M_{14} = -7$$

$$M = \begin{bmatrix} -4 & -2 & -7 & -6 \\ -3 & -1 & 0 & 0 \\ 0 & 0 & -1 & -1 \\ 1 & 0 & -8 & -7 \end{bmatrix}$$

Find Cofactors:-

$$C_{11} = (-1)^{1+1} M_{11} = (+1)(-4) = -4$$

$$C_{12} = (-1)^{1+2} M_{12} = (-1)(0) = 0$$

$$C_{13} = (-1)^{1+3} M_{13} = (+1)(-8) = -8$$

$$C_{14} = (-1)^{1+4} M_{14} = (-1)(-7) = 7$$

$$C_{21} = (-1)^{2+1} M_{21} = (-1)(-3) = 3$$

$$C_{22} = (-1)^{2+2} M_{22} = (+1)(-1) = -1$$

$$C_{23} = (-1)^{2+3} M_{23} = (-1)(0) = 0$$

$$C_{24} = (-1)^{2+4} M_{24} = (+1)(0) = 0$$

$$C_{31} = (-1)^{3+1} M_{31} = (+1)(0) = 0$$

$$C_{32} = (-1)^{3+2} M_{32} = (-1)(0) = 0$$

$$C_{33} = (-1)^{3+3} M_{33} = (+1)(-1) = -1$$

$$C_{34} = (-1)^{3+4} M_{34} = (-1)(-1) = 1$$

$$C_{41} = (-1)^{4+1} M_{41} = (-1)(1) = -1$$

$$C_{42} = (-1)^{4+2} M_{42} = (+1)(0) = 0$$

$$C_{43} = (-1)^{4+3} M_{43} = (-1)(-8) = 8$$

$$C_{44} = (-1)^{4+4} M_{44} = (+1)(-7) = -7$$

$$C = \begin{bmatrix} -4 & 2 & -7 & 6 \\ 3 & -1 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ -1 & 0 & 8 & -7 \end{bmatrix}$$

$$C^T = \begin{bmatrix} -4 & 3 & 0 & -1 \\ 2 & -1 & 0 & 0 \\ -7 & 0 & -1 & 8 \\ 6 & 0 & 1 & -7 \end{bmatrix}$$

→ adj(A)

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

$$A^{-1} = \frac{1}{1} \begin{bmatrix} -4 & 3 & 0 & -1 \\ 2 & -1 & 0 & 0 \\ -7 & 0 & -1 & 8 \\ 6 & 0 & 1 & -7 \end{bmatrix}$$

→ Ans.

19-22-

Find $\det(A)$ if zero leave it there
but if not 0 then use some method
to find inverse.

Question: 24-29

28-

$$-x_1 - 4x_2 + 2x_3 + x_4 = -32$$

$$2x_1 - x_2 + 7x_3 + 9x_4 = 14$$

$$-x_1 + x_2 + 3x_3 + x_4 = 11$$

$$x_1 - 2x_2 + x_3 - 4x_4 = -4$$

$$\begin{bmatrix} -1 & -4 & 2 & 1 \\ 2 & -1 & 7 & 9 \\ -1 & 1 & 3 & 1 \\ 1 & -2 & 1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -32 \\ 14 \\ 11 \\ -4 \end{bmatrix}$$

$$A \quad x \quad B$$

$$x = A^{-1}B$$

$$A_1 = \begin{bmatrix} -32 & -4 & 2 & 1 \\ 2 & -1 & 7 & 9 \\ -1 & 1 & 3 & 1 \\ 1 & -2 & 1 & -4 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} -1 & -32 & 2 & 1 \\ 2 & 14 & 7 & 9 \\ -1 & 11 & 3 & 1 \\ 1 & -4 & 1 & -4 \end{bmatrix}$$

EXERCISES

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$$A_3 = \begin{bmatrix} -1 & -4 & -32 & 1 \\ 2 & -1 & 14 & 9 \\ -1 & 1 & 11 & 1 \\ 1 & -2 & -4 & -4 \end{bmatrix}, A_4 = \begin{bmatrix} -1 & -4 & 2 & -32 \\ 2 & -1 & 7 & 14 \\ -1 & 1 & 3 & 11 \\ 1 & -2 & 1 & -4 \end{bmatrix}$$

$$\det(A) = ?$$

$$\det(A) = \begin{vmatrix} -1 & -4 & 2 & 1 \\ 2 & -1 & 7 & 9 \\ -1 & 1 & 3 & 1 \\ 1 & -2 & 1 & -4 \end{vmatrix}$$

$$\det(A) = \begin{vmatrix} 0 & -6 & 3 & -3 \\ 0 & 3 & 5 & 17 \\ 0 & -1 & 4 & -3 \\ 1 & -2 & 1 & -4 \end{vmatrix} \begin{array}{l} R_3 + R_4 \\ R_2 - 2R_4 \\ R_1 + R_4 \end{array}$$

Expand along 1st col

$$\det(A) = 0 + 0 + 0 + (-1)(1) \begin{vmatrix} -6 & 3 & -3 \\ 3 & 5 & 17 \\ -1 & 4 & -3 \end{vmatrix}$$

$$\det(A) = -1 \begin{vmatrix} -6 & 3 & -3 \\ 3 & 5 & 17 \\ -1 & 4 & -3 \end{vmatrix}$$

expand along 1st row

$$\det(A) = -1 \begin{vmatrix} -6 & 5 & 17 & -3 \\ 3 & 17 & -3 & 3 \\ 4 & -3 & -1 & -3 \\ -1 & 4 & -3 & -1 \end{vmatrix}$$

$$\det(A) = -1 \left[-6(-15-18) - 3(-9+17) - 3(42+5) \right]$$

$$\det(A) = - (498 - 24 - 51)$$

$$\det(A) = -423 \text{ Ans}$$

$$\det(A_1) = ?$$

$$\det(A_1) = \begin{vmatrix} -32 & -4 & 2 & 1 \\ 14 & -1 & 7 & 9 \\ 11 & 1 & 3 & 1 \\ -4 & -2 & 1 & -4 \end{vmatrix}$$

$$\det(A_1) = \begin{vmatrix} 12 & 0 & 14 & 5 \\ 25 & 0 & 10 & 10 \\ 11 & 1 & 3 & 1 \\ 18 & 0 & 7 & -2 \end{vmatrix} \begin{array}{l} \\ R_1 + 2R_3 \\ R_2 + R_3 \\ R_1 + 4R_3 \end{array}$$

Expand along C_2

$$\det(A_1) = 0 + 0 + (-1)^{3+2} \begin{vmatrix} 12 & 14 & 5 \\ 25 & 10 & 10 \\ 18 & 7 & -2 \end{vmatrix} + 0$$

$$\det(A_1) = -1 \begin{vmatrix} 12 & 14 & 5 \\ 25 & 10 & 10 \\ 18 & 7 & -2 \end{vmatrix}$$

expand along R_1

$$\det(A_1) = -1 \begin{vmatrix} 12 & 10 & 10 & 1 \\ 7 & -2 & -14 & 25 \\ 18 & -2 & 18 & 7 \end{vmatrix}$$

$$\det(A_1) = -1 [12(-20 - 70) - 14(-30 - 180) + 5(115 - 180)]$$

$$\det(A_1) = -1 (-1080 + 3270 - 25)$$

$$\det(A_1) = -2115$$

$$\det(A_2) = ?$$

$$\det(A_2) = \begin{vmatrix} -1 & -32 & 2 & 1 \\ 2 & 14 & 7 & 9 \\ -1 & 11 & 3 & 1 \\ 1 & -4 & 1 & -4 \end{vmatrix}$$

$$\det(A) = \begin{vmatrix} 0 & -36 & 3 & -3 \\ 0 & 22 & 5 & 17 \\ 0 & 7 & 4 & -3 \\ 1 & -4 & 1 & -4 \end{vmatrix} \begin{array}{l} R_3 + R_4 \\ R_2 - 2R_4 \\ R_1 + R_4 \end{array}$$

Expand along 1st col.

$$\det(A) = 0 + 0 + 0 + (-1)^{4+1} (1) \begin{vmatrix} 36 & 3 & -3 \\ 22 & 5 & 17 \\ 7 & 4 & -3 \end{vmatrix}$$

$$\det(A) = -1 \begin{vmatrix} -36 & 3 & -3 \\ 22 & 5 & 17 \\ 7 & 4 & -3 \end{vmatrix}$$

expand along R_1

$$\det(A_2) = -1 \begin{vmatrix} -36 & 5 & 17 & -3 & 22 & 17 & -3 & 22 & 5 \\ & 4 & -3 & & 7 & -3 & & 7 & 4 \end{vmatrix}$$

$$\det(A_2) = -1 \left[-36(-15-68) - 3(-66-119) - 3(88-35) \right]$$

$$\det(A_2) = -1 \cdot (+2988 + 555 - 159)$$

$$\det(A_2) = -3384$$

$$\det(A_3) = ?$$

$$\det(A_3) = \begin{vmatrix} -1 & -4 & -32 & 1 \\ 2 & -1 & 14 & 9 \\ -1 & 1 & 11 & 1 \\ 1 & -2 & -4 & -4 \end{vmatrix}$$

$$\det(A_3) = \begin{vmatrix} 0 & -6 & -36 & -3 \\ 0 & 3 & 22 & 17 \\ 0 & -1 & 7 & -3 \\ 1 & -2 & -4 & -4 \end{vmatrix} \begin{array}{l} R_3 + R_4 \\ R_2 - 2R_4 \\ R_1 + R_4 \end{array}$$

Expand along C_1

$$\det(R_3) = 0 + 0 + 0 + (-1)(1) \begin{vmatrix} -6 & -36 & -3 \\ 3 & 22 & 17 \\ -1 & 7 & -3 \end{vmatrix}$$

$$\det(A_3) = -1 \begin{vmatrix} -6 & -36 & -3 \\ 3 & 22 & 17 \\ -1 & 7 & -3 \end{vmatrix}$$

expand along R_1

$$\det(A_3) = -1 \left[-6 \begin{vmatrix} 22 & 17 \\ 7 & -3 \end{vmatrix} + 36 \begin{vmatrix} 3 & 17 \\ -1 & -3 \end{vmatrix} - 3 \begin{vmatrix} 3 & 22 \\ -1 & 7 \end{vmatrix} \right]$$

$$\det(A_3) = -1 \left[-6(-66 - 119) + 36(-9 + 17) - 3(21 + 22) \right]$$

$$\det(A_3) = -1 \left[1110 + 228 - 129 \right]$$

$$\det(A_3) = 1269$$

$$\det(A_4) = ?$$

$$\det(A_4) = \begin{vmatrix} -1 & -4 & +2 & -32 \\ 2 & -1 & 7 & 14 \\ -1 & 1 & 3 & 11 \\ -1 & -2 & 1 & -4 \end{vmatrix}$$

$$\det(A_4) = \begin{vmatrix} 0 & -6 & +3 & -36 \\ 0 & 3 & 5 & 22 \\ 0 & -1 & 4 & 7 \\ 1 & -2 & 1 & -4 \end{vmatrix} \begin{array}{l} R_3 + R_4 \\ R_2 - 2R_4 \\ R_1 + R_4 \end{array}$$

expand along C_1

$$\det(A_4) = 0 + 0 + 0 + (-1)^{4+1} (1) \begin{vmatrix} -6 & +3 & -36 \\ 3 & 5 & 22 \\ -1 & 4 & 7 \end{vmatrix}$$

$$\det(A_4) = -1 \begin{vmatrix} -6 & 3 & -36 \\ 3 & 5 & 22 \\ -1 & 4 & 7 \end{vmatrix}$$

expand along R_1

$$\det(A_4) = -1 \left[-6 \begin{vmatrix} 5 & 22 \\ 4 & 7 \end{vmatrix} - 3 \begin{vmatrix} 3 & 22 \\ -1 & 7 \end{vmatrix} - 36 \begin{vmatrix} 3 & 5 \\ -1 & 4 \end{vmatrix} \right]$$

$$\det(A_4) = -1 \left[-6(35 - 88) - 3(21 + 22) - 36(12 + 5) \right]$$

$$\det(A_4) = -1 (318 - 129 - 612)$$

$$\det(A_4) = -1 (-423)$$

$$\det(A_4) = 423$$

$$x_1 = ?$$

$$x_1 = \frac{|A_1|}{|A|} = \frac{-2115}{-423}$$

$$x_1 = 5$$

$$x_2 = ?$$

$$x_2 = \frac{|A_2|}{|A|}$$

$$x_2 = \frac{-3384}{-423}$$

$$x_2 = 8$$

$$x_3 = ?$$

$$x_3 = \frac{|A_3|}{|A|} = \frac{-1269}{-423}$$

$$x_3 = 3$$

$$x_4 = ?$$

$$x_4 = \frac{|A_4|}{|A|} = \frac{423}{-423}$$

$$x_4 = -1$$

24-27, 29

The matrix having size less than or equal to 3×3 then solve with normal method otherwise follow method of q: 28 to find determinant

Question 230-

Show that matrix

$$A = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

is invertible for all values of θ then find A^{-1} using Theorem

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

$\det(A) = 1$ $\rightarrow$ Since 1 so always invertible

Note: Use method of q: 23 to solve for inverse.

Question : 33

Let

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Assuming $\det(A) = -7$ find.

a-

$$\det(3A)$$

$$\det(3A) = \det(3A)$$

$$\det(3A) = (3)^3 \det(A) \quad \therefore \det(kA) = k^n \det(A)$$

$$\det(3A) = 27(-7)$$

$$\det(3A) = -189 \text{ Ans}$$

b-

$$\det(A^{-1})$$

$$\det(A^{-1}) = \det(A^{-1})$$

$$\det(A^{-1}) = \frac{1}{\det(A)} \quad \therefore \det(A^{-1}) = \frac{1}{\det(A)}$$

$$\det(A^{-1}) = -\frac{1}{7} \text{ Ans}$$

c-

$$\det(2A^{-1})$$

$$\det(2A^{-1}) = \det(2A^{-1})$$

$$\det(2A^{-1}) = (2)^3 \det(A^{-1}) \quad \therefore \det(kA) = k^n \det(A)$$

$$\det(2A^{-1}) = \frac{8}{\det(A)} \quad \therefore \det(A^{-1}) = \frac{1}{\det(A)}$$

$$\det(2A^{-1}) = -\frac{8}{7} \text{ Ans}$$

d-

$$\det((2A)^{-1})$$

$$\det(2A)^{-1} = \frac{\det(2A)}{\det(A)}$$

$$\det(2A)^{-1} = \frac{1}{\det(2A)}$$

$$\therefore \det(A^{-1}) = \frac{1}{\det(A)}$$

$$\det(2A)$$

$$\det(A)$$

$$\det(2A)^{-1} = \frac{1}{(2)^3 \det(A)}$$

$$\therefore \det(kA) = k^n \det(A)$$

$$(2)^3 \det(A)$$

$$\det(2A)^{-1} = -\frac{1}{56} \text{ Ans}$$

e-

$$\det \begin{bmatrix} a & g & d \\ b & h & e \\ c & i & f \end{bmatrix}$$

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Take Transpose of A

$$\begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$$

$\Rightarrow$

$$\begin{bmatrix} a & g & d \\ b & h & e \\ c & i & f \end{bmatrix}$$

$C_1 \leftrightarrow C_3$

$$\text{As we know } \det(A^T) = \det(A) = 7$$

$$\text{and also by Swapping } \det = -\det(A^T)$$

$$\text{So } \det = -(-7) = 7 \text{ Ans}$$

Question: 34

In each part ; Find determinant given
A is 4×4 matrix for which $\det(A) = -2$

a-

$$\det(-A)$$

$$\det(-A) = \det(-A)$$

$$\text{Using rule } k = -1 \quad \therefore \det(kA) = k^n \det(A)$$

$$\det(-A) = (-1)^4 \det(A)$$

$$\det(-A) = 1(-2) = -2 \text{ Ans.}$$

b-

$$\det(A^{-1})$$

$$\det(A^{-1}) = \det(A^{-1}) \quad \therefore \det(A^{-1}) = \frac{1}{\det(A)}$$

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

$$\det(A^{-1}) = -\frac{1}{2} \text{ Ans.}$$

c-

$$\det(2A^T)$$

$$\det(2A^T) = \det(2A^T)$$

$$\det(2A^T) = (2)^4 \det(A^T) \quad \therefore \det(kA) = k^n \det(A)$$

$$\det(2A^T) = 16 \det(A) \quad \therefore \det(A^T) = \det(A)$$

$$\det(2A^T) = 16(-2)$$

$$\det(2A^T) = -32 \text{ Ans.}$$

d-

$$\det(A^3)$$

$$\det(A^3) = \det(A^3)$$

$$\det(A^3) = (\det(A))^3 \quad \therefore \det(A^k) = (\det(A))^k$$

$$\det(A^3) = (-2)^3$$

$$\det(A^3) = -8 \text{ Ans}$$

Question : 35

In each part, find determinant given that A is 3×3 matrix for which $\det(A) = 7$

a-

$$\det(3A)$$

$$\det(3A) = \det(3A)$$

$$\det(3A) = (3)^3 \det(A) \quad \therefore \det(kA) = k^n \det(A)$$

$$\det(3A) = 27(7)$$

$$\det(3A) = 189$$

b-

$$\det(A^{-1})$$

$$\det(A^{-1}) = \det(A^{-1})$$

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

$$\therefore \det(A^{-1}) = \frac{1}{\det(A)}$$

$$\det(A)$$

$$\det(A)$$

$$\det(A^{-1}) = \frac{1}{7} \text{ Ans}$$

c-

$$\det(2A^{-1})$$

$$\det(2A^{-1}) = \det(2A^{-1})$$

$$\det(2A^{-1}) = (2)^3 \det(A^{-1}) \quad \therefore \det(kA) = k^n \det(A)$$

$$\det(2A^{-1}) = \frac{8 \cdot 1}{\det(A)} \quad \therefore \det(A^{-1}) = \frac{1}{\det(A)}$$

$$\det(2A^{-1}) = \frac{8}{1} \text{ Ans}$$

d-

$$\det(2A)^{-1}$$

$$\det(2A)^{-1} = \det((2A)^{-1})$$

$$\det((2A)^{-1}) = \frac{1}{\det(2A)} \quad \therefore \det(A^{-1}) = \frac{1}{\det(A)}$$

$$\det((2A)^{-1}) = \frac{1}{(2)^3 \det(A)} \quad \therefore \det(kA) = k^n \det(A)$$

$$\det((2A)^{-1}) = \frac{1}{8 \times 7} = \frac{1}{56} \text{ Ans}$$